

Lab Report Writing Instructions

- There will be one lab report from each group every week.
- Lab Report must be **HANDWRITTEN**.
- Use proper **margin** (using ruler) on the left side of papers.
- Write on both sides of an **A4 paper**.
- The separate sections must be **CLEARLY LABELLED** and in **PROPER ORDER**.
- Use the data obtained (measured) for your lab report.
- All data tables, circuit drawings must be done neatly and labeled properly.
- Use proper **A4 size graph paper** for any graphs needed and label everything accordingly.
- **DO NOT PLAGIARIZE OR COPY OTHER'S REPORT.**

1. **Cover Page:** All lab reports should have a cover page, and the same cover page should be used for all the lab reports. Use the given cover page format in CANVAS and add the necessary information there. You have to add names and IDs of all group members and must mention the writer of the report.
2. **Experiment Name:** Write the experiment name clearly on the Cover Page.
3. **Objective:** You should briefly write what the aim of the experiment was. In other words, write what you intend to achieve by conducting the experiment.
4. **Theory:** In this section of the Lab Report, you will specifically write only the things taught during the lab. This section should be concise, to the point, and in your own words. Direct copying from the lab manual/other groups/any website is strictly prohibited. Marks will be given based on your ability to explain what you understood during class.
5. **List of Equipment:** A simple list of all the apparatuses and Equipment you used to do the lab experiment.
6. **Circuit Diagram:** Draw the circuit diagram neatly and label properly.
7. **Results/Data/Graphs:** This section of the lab report will contain the data that you have collected practically, and it should be presented in a tabular form (make a clean table of the data obtained in class). Detailed calculations must be included here. If you have to calculate any relevant percentage error or percentage difference from the practical data, you should put those in this section. Add graphs to represent your data and results if required. The graph must be hand-drawn on graph paper with proper labels and scale on both axes. Try to use at least 70% of the graph instead of using a small section of the graph paper with a small vertical and horizontal scale.
8. **Q/A Section**
9. **Result Analysis and Discussion:** This is one of the most important parts of the lab report. What you write here proves how attentive and careful you were during the lab class. Copying a single line from another person's discussion or from a previous lab report will earn you a straight zero. **In this section, you have to analyze and explain all your data, results, and graphs.** Your analysis will be based on what type of data/result and graph you are expecting (from your theory knowledge) and what type of data/result and graph you are getting from the experiment. Please mention any factors for discrepancies between these two.
10. **Conclusion:** Here, you have to summarize the findings of this experiment based on the concepts you learned from theory and the outcomes/lessons you got from the experiment. If you have any opinion or suggestion on the experiment, you can mention that too.
11. **Reference:** Site/mention your source of information. Try to follow IEEE reference guide.

*****Attachment:** Finally, you must attach the signed datasheet

- **Multisim Simulation:** Take a screenshot of your simulation circuit along with the result table and graph, and paste everything into a Word document and submit through CANVAS along with the Multisim file. You must mention your name, ID, and Group number on both files.